# Human Patience and AI Frustration:
# How Subsequent Chatbot Mistakes Affect Users Trust, Attention, and Behavior

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Google Forms survey results

**Abstract**

AI Chatbots have been used a lot more in education, workplace, and problem-solving in daily life activities. Nevertheless, repeated errors might affect the way users behave when dealing with such systems. This study investigates and examines the effect of repeated AI chatbot errors on the participants' frustration, trust, reading attention, and behaviors within scenario-based surveys. Altogether 204 responses have been collected and tools in google sheets such as content duplication tests were employed to filter out approximately 181 clean responses. The trust in AI, frustration at different levels of repetition of the chatbot error in two scenarios, changes in reading attention following the chatbot error, and the behavior after repeated errors have been assessed. Repeated errors have increased the frustration, which means that the mean frustration of Scenario A increased from 2.93 for one incorrect answer to 5.77 for eight incorrect answers and Scenario B changed from 3.34 early to 5.59 close to the end of the scenario. Participants had moderate expectations for the correctness of AI (mean = 4.14) but lower trust in it (mean = 3.56). Importantly, repeated errors did not lead to giving up by users; many respondents have claimed increased reading attentiveness, reformulating of their questions, and switching to alternatives and different google/search engines.

## Introduction

AI chatbots are becoming a huge part of our daily digital experience and shapes the way we tackle issues and problems. Students use these tools to get assistance with their studies, generate ideas, and write. Even adults and employees use them to become more productive; and regular users turn to them for summarizations, workout routines, and other general suggestions. The fluency of chatbot responses leads to users regarding them as valuable resources regardless of their lack of certainty about the validity of the provided answers. Thus, one encounters an important human-AI interaction issue. What to do when a seemingly helpful resource provides wrong answers over and over again. Previous work on large language models has revealed that fluent responses could be unjustified, misguiding, and factually wrong (Bender et al., 2021; Huang et al., 2023).

When it comes to AI chatbots making a mistake, a single error may be annoying but is not a huge problem for most users and is usually swept away. On the other hand, repetition of errors is different. A repetition of errors can affect how the whole process unfolds and what decision the user makes: to continue relying on the bot, to ask again or to check the answer, to use some other software or to give up on the task altogether. In this experiment, the repetition of mistakes made by AI is viewed as a problem of frustration as well as a problem of behavior decisions.

The errors made by AI systems or chatbots are very different in nature compared to those committed by a typical conventional software. While a button or page freeze occurs through a detectable and predictable error pattern, an AI assistant such as a chatbot may correctly answer one question and then provide an incorrect response to the very next request posed by the user. This unpredictability makes it difficult to gauge the level of trust in the system since users will have no clear idea when the AI system will provide accurate information. What is more, the core concern lies not only in whether the chatbot provides an inaccurate response, but whether the user can predict that it will make a mistake.

This research paper mainly concentrates on regular users of AI and tries to determine how the constant errors of chatbots influence users frustration, trust, attention spans, and behavioral intentions, such as entirely giving up or switching to another tool. The primary thesis supported by the results of the survey is that the errors of AI cannot be reduced only to creating frustration. These repeated mistakes from AI chatbots could prompt the user to read its responses more carefully, rephrase prompts, or check information using other alternatives.

## Research Question and Hypotheses

Research Question: What is the effect of repeated AI chatbots mistakes on user frustration, trust, reading attention, and behavior such as rephrasing, switching tools, or giving up?

- H1: As the number of AI chatbot mistakes increases, the frustration levels of users will also increase.
- H2: Inconsistent AI mistakes will induce similar or greater frustration than consistently wrong mistakes since users cannot easily determine when the AI can be trusted.
- H3: Greater frustration will be correlated with self-reports of changes in reading attention.
- H4: Greater frustration will be correlated with intentions to drop out or switch away, which can be assessed by user reports of quitting, switching to a different AI, or switching to Google.

## Literature Review

### Human-Computer Frustration and Repeated Errors

From research conducted into human-computer interaction, it is clear that frustration occurs when technology gets in the way of user objectives, wastes time, or causes the user to perform additional tasks for recovery purposes. As stated by Ceaparu et al. (2004) in their study into end-user frustration, computer problems are not simply technological in nature but emotional as well because computer problems cause people to waste time and effort when they try to achieve an objective. Similarly, Lazar et al. (2006) state that computer frustration may have tangible effects on users since computer frustrations may include blocking or causing users to repeat tasks. This is relevant to chatbot design since the incorrect response from a chatbot will not just indicate one erroneous output.

These previous studies that examined HCI explained why repetitive errors of AI can be more harmful than one error alone. While, in case of a software error, the user will notice that there is an error made by the software and perhaps know why it occurred. In case of a mistake made by

AI during a conversation with the bot, the error will appear within the sentence which may look correct. It will be really hard to find the mistake and can lead to an unpleasant result. People can ask help with an issue about which they know nothing from AI. But since the bot replied incorrectly, people will use the reply regardless of its response. Therefore, the frustration during a conversation with the bot will not only result from the mistake but trying to establish whether it is useful or not.

## Trust in Automation and Trust Calibration

"Trust" is one of the most important concepts that connect the areas of automation research and AI chatbot implementation. According to Lee & See (2004), trust in automation becomes critical for users because they have to use automated tools in those cases when it is useful and not use them in those cases when their actions are incorrect. It means that the purpose is not the maximal level of trust, but the appropriate level of trust. Problems of misuse, disuse, and abuse of automation were introduced by Parasuraman & Riley (1997). In other words, there is an opportunity for excessive reliance on automation, rejection of automation, and misuse of automation. For AI chatbot implementation, it means that the user's attitude towards errors will depend on the ability of the chatbot to help in calibration of trust.

The current research also relates to algorithm aversion. Dietvorst et al. (2015) have established that humans tend to reject algorithms' recommendations after making a mistake and regardless of how well the algorithm may work on average. It may prove helpful for analyzing chatbot frustration because the user will not just evaluate the quality of the interaction based on the overall average value of the algorithm. On the contrary, the error will greatly diminish the subsequent trust in the algorithm. In this case, the problem is reflected in the question about trust reduction and the first action question because several respondents started using Google or another website after receiving several incorrect responses from the chatbots.

## Cognitive Load and Verification Effort

The "Cognitive Load Theory" provides us another framework to interpret how chatbot errors may frustrate users. According to Sweller (1988), there are limitations on the working memory capacity that humans possess during problem-solving and that learning or performing the task will be adversely affected by any additional cognitive load which is unnecessary. Mayer (2005) further pointed out the need to make information designs such that the user's attention is devoted to only the primary task. In terms of artificial intelligence chatbots, the additional cognitive load stems from the fact that the user will have to evaluate the answer and verify its correctness.

This concept as described, directly relates and connects to the theory of the verification shift as proposed in this research paper. The user needs to concentrate only on the task at hand if the provided AI-based response is trustworthy. If it is potentially unreliable, the user should concentrate on both the task and the reliability of the tool. This additional load may serve to explain why the subjects in the current experiment tended to read the text more carefully after making mistakes. Although it may seem like an increased level of attention, increased verification due to decreased trust is another possible interpretation of it.

### AI Reliability, Hallucination, and Fluent Wrong Answers

The latest AI chatbots are based on language models capable of generating fluent texts similar to human writing. Yet, studies on language models have found several times that fluent language does not necessarily mean accurate facts. According to Bender et al., (2021), large language models can create fluent language but lack grounded knowledge. As a result, people may take this information as something more reliable than it is supposed to be. The latest studies on large language models' hallucinations show how such models may generate plausible but non-factual information, particularly, in case an answer requires factual grounding or goes beyond the area where it is reliable (Huang et al., 2023; Ji et al., 2023).

Hallucination research is relevant for the current research paper since it describes the reasons why errors made by AI are hard for users to deal with. A response made by a chatbot can be wrong and at the same time be confident, polite, and fluent, so it's hard for users to differentiate on what is right or wrong. Therefore, there is an altered user experience compared to searching information online via engines that offer sources or various links. In case when a chatbot gives a single response, users may have to find sources themselves. It all fits into the topic of verification behaviors that users tend to display after errors are made.

### Crowdsourcing, Online Survey Behavior, and the Current Gap

Due to the fact that this paper used an online Google Form survey, it is also necessary to mention literature on online samples and crowdsourcing participation. Kittur et al. (2008) demonstrated that online labor markets could be used for user studies, but it also means that researchers should pay more attention to participant attention and data quality. Gadiraju et al. (2015) found out that online studies could include low-effort or malicious participation, which means that data filtering is crucial. It is necessary to discuss this point because this paper used a convenience sample and found duplicate content among participants later on. To remove duplicate content among participants, google sheet tools were used.

Ultimately, the literature shows that the experience of repetitive errors of an AI chatbot is related to several topics of research in psychology and HCI including frustration with technology, trust in automation, cognitive load, algorithm aversion, and reliability of AI. Current research explains the nature of frustration with technology errors, the need for proper calibration of trust, and the fact that fluency in answers from an AI program does not mean that it provides reliable results. Still, there is limited information regarding the ways in which users perceive their actions and behaviors in case of repetitive mistakes of a chatbot.

# Method

## Design

The research design applied to this study was cross-sectional, scenario-based. Participants took one Google Forms survey where they provided their information regarding AI usage, their trust towards the AI chatbots, their frustration when faced with repeated errors, change in attention and potential actions they would take after experiencing repeated wrong answers. The fact that all participants went through Scenario A and Scenario B reveals that the study can be considered as a within-subjects scenario comparison study embedded in a cross-sectional survey, not a live chatbot experiment where real life experiments would take place.

## Participants

There were 204 submitted surveys. During the data collection process, 23 duplicates based on content were found, and there were 181 unique patterns of responses left for cleaned descriptive analysis. To clear out these duplicate responses, we used built-in google sheet tools. This sample was a non-randomly recruited convenience sample of online and school/social-network respondents who had experience with AI chatbots.

**Table 1**
**Participant and sample summary**

| Measure | Result |
|---|---|
| Submitted responses | 204 |
| Cleaned unique response patterns | 181 |
| Multiple-times-a-day AI users | 62 (34.3%) |
| Intermediate AI users | 97 (53.6%) |
| Most common age group | 25-34 |

## Materials

The survey was titled as "AI Interaction Experience Survey" and was sent out via Google Forms. In order to share the survey and to gather responses, the google form was shared in media such as Reddit, SurveyJunkie, and to close members. It contained AI interaction questions, trust and accuracy expectations, two error repetition scenarios, attention and behavior questions, a modified frustration scale of five items, general reliability and consistency expectations, a voluntary open-ended response, and demographic questions.

## Scenario Conditions

In Scenario A, there was a scenario involving consistent AI failure. The participants envisioned themselves interacting with an AI in search of 10 facts, receiving incorrect responses at each step of the way. Frustration ratings for one incorrect answer, three incorrect answers, five incorrect answers, and eight incorrect answers were asked for.

Scenario B involved inconsistent AI interaction. The participants envisioned themselves interacting with an AI that gave correct responses to certain questions and wrong responses to others, leaving them unsure as to which answers to believe in. Frustration ratings were collected early on in the process, halfway, and close to the end.

## Measures

AI usage at baseline was assessed by usage of the chatbot platform, usage frequency, and experience level with AI. Trust was assessed by factual accuracy, general trust, and trust reduction due to an incorrect response, on a 1-7 scale. Frustration trajectory was assessed through the Scenario A grid and Scenario B grid. Attention was assessed by reading behaviors and a threshold for shifting attention. Intentions to drop out or switch away was assessed with a first action question following wrong responses. A main frustration scale was developed based on five items regarding giving up, time use, mental drain, confidence, and mental effort. The positive items were reversed prior to averaging. Reliability and consistency were also measured on a 1-7 scale.

## Procedure

Participants opened the Google form, reviewed the instructions and questions, completed questions that assessed for AI chatbot usage, completed the trust and expected accuracy measures, followed by the two scenarios with wrong responses, followed by the attention and behavior measure, then completed the adapted frustration scale, reliability and consistency measures, and finished with demographic questions.

## Data Cleaning and Analysis Plan

This research used multiple tools such as descriptive statistical analysis and the examination of users' responses. We reviewed users' responses using mean, frequency, and percentage in order to discover the patterns of frustration, trust, attention, and behavior. For categorical questions, we calculated the number and percentages; for questions measured on 1-7 scale, mean and standard deviation were computed. Frustration trajectory was analyzed based on a comparison of users' frustration on the scenario checkpoints. Switch-away behavior was evaluated by combining responses of "switch to Google/search," "switch to a different AI," and "give up on the task" into the switch-away category and "rephrase/simplify" and "retry with the same question" into stay-and-repair responses. Responses duplication and submission effort could be present in online surveys. Hence, the patterns of duplication should be identified prior to any descriptive analysis of the data (Gadiraju et al., 2015; Kittur et al., 2008). The optional written responses were analyzed to reveal themes explaining quantitative findings.

# Results

## Participant Characteristics

In total, after cleaning the dataset, there were about 181  unique responses. After reviewing our data, we can see that the majority of participants used AI technologies regularly. From our data, 62 participants (34.3%) used AI technologies several times per day, and 51 participants (28.2%) a few times per week. Most participants regarded themselves as intermediate users (97 participants, 53.6%), then beginner users (44 participants, 24.3%), and advanced users (40 participants, 22.1%). ChatGPT Free was the most popular chatbot platform among participants (97 participants, 53.6%), then Gemini (38 participants, 21.0%), and Claude (37 participants, 20.4%).

**Table 2**
**Chatbot platforms selected by respondents**

| Category | n | % |
|---|---|---|
| ChatGPT (Free) | 97 | 53.6 |
| Gemini | 38 | 21.0 |
| Claude | 37 | 20.4 |
| ChatGPT (Plus) | 30 | 16.6 |
| Other | 25 | 13.8 |
| Copilot | 25 | 13.8 |

## Baseline Trust and Accuracy Expectations

Moderately accurate responses were expected, meaning users somewhat trusted AI chatbots accuracy, ($M = 4.14$, $SD = 1.56$) although overall trust level was lower ($M = 3.56$, $SD = 1.62$). The drop in trust level after an incorrect response was found to be high ($M = 4.50$, $SD = 1.77$). Overall rating of general reliability was lower ($M = 3.39$, $SD = 1.47$), but the rating of consistency or predictability was higher ($M = 4.06$, $SD = 1.34$).

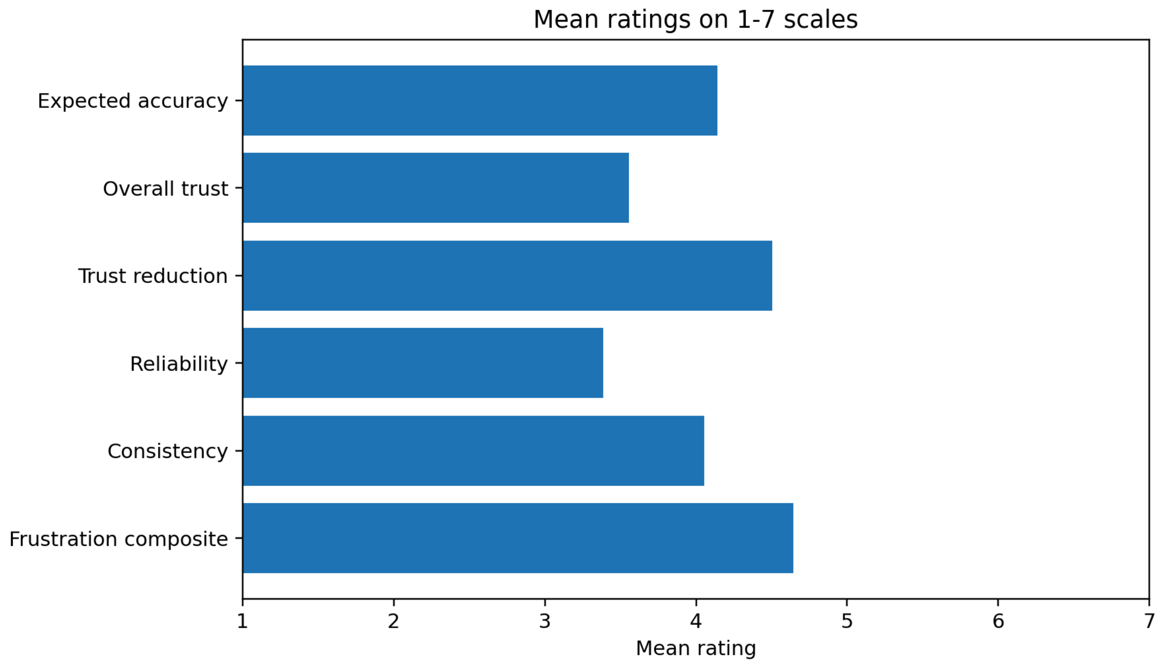


*Figure 1. Mean ratings for trust, reliability, consistency, and frustration-related measures.*

## Frustration Increased Across Repeated Errors

In both scenarios, frustration levels were seen to increase as error rates accumulated. In Scenario A, the average level of frustration increased from 2.93 (SD = 1.72) when the first wrong answer occurred to 5.77 (SD = 1.68) after eight wrong answers. In Scenario B, the average level of frustration increased from 3.34 (SD = 1.58) early in the interaction to 5.59 (SD = 1.52) later on in the interaction.

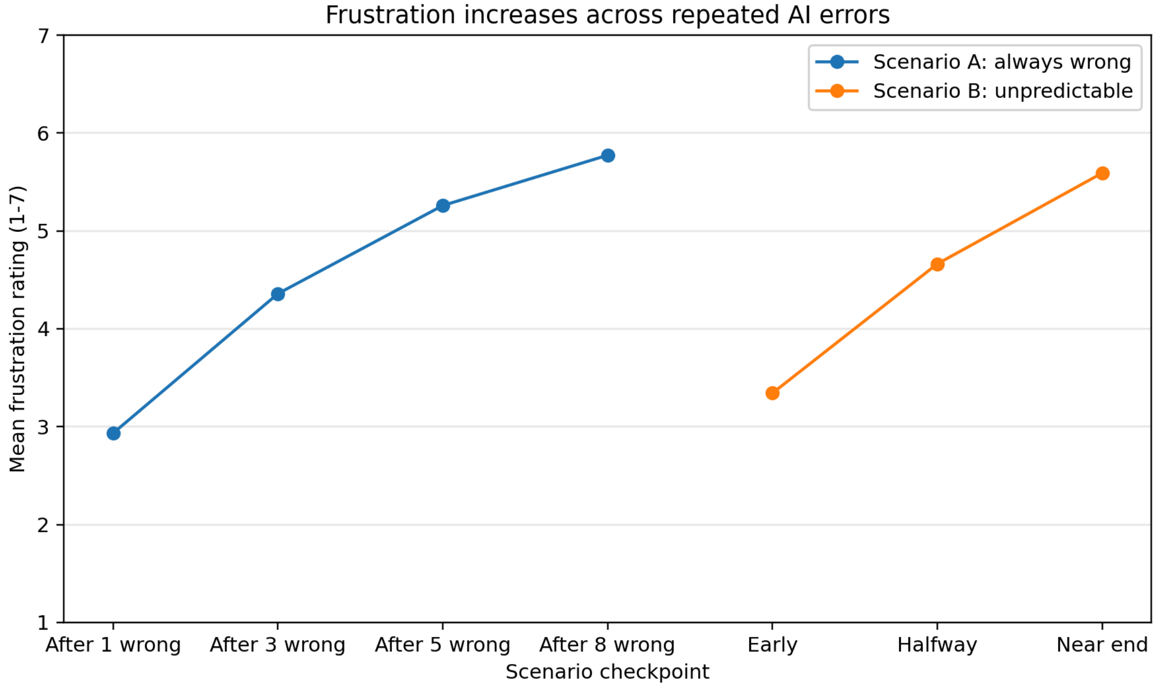


*Figure 2. Mean frustration ratings across repeated-error checkpoints in Scenario A and Scenario B.*

**Table 3**
**Mean frustration ratings by scenario checkpoint**

| Scenario | Checkpoint 1 | Checkpoint 2 | Checkpoint 3 | Final checkpoint |
|---|---|---|---|---|
| A: always wrong | 2.93 | 4.36 | 5.26 | 5.77 |
| B: unpredictable | 3.34 | 4.66 | 5.59 | N/A |

## Consistent Versus Inconsistent Failure

When participants were asked about their preference regarding types of failure that frustrated them more, the majority of 75 respondents (41.4%) said “both equally”. The number of participants who chose Scenario B (unpredictable error) was 60 (33.1%), while those who chose Scenario A (always-wrong errors) were 35 (19.3%). Only 11 (6.1%) participants chose neither.

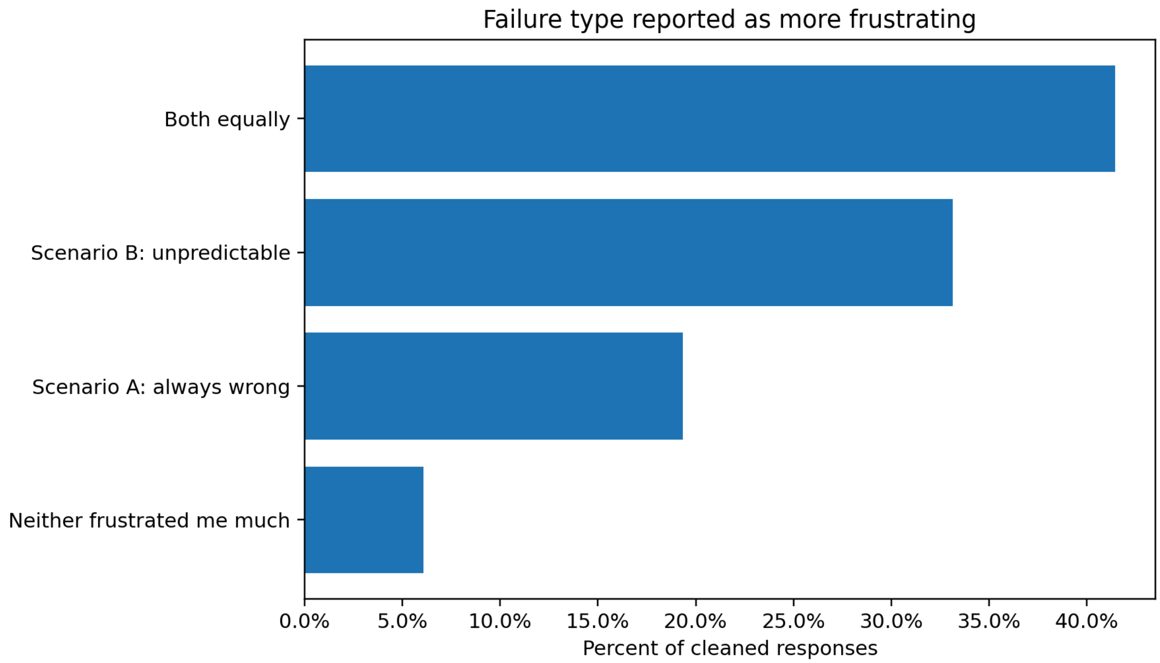


*Figure 3. Failure type reported as more frustrating overall.*

## Attention Shift After AI Mistakes

The most popular choice that participants made was that repeating mistakes made them pay closer attention to reading AI responses (106, 58.6%). Fewer people reported skimming AI responses much less carefully (29, 16.0%), slightly less carefully (19, 10.5%), or not changing their way of interaction at all (27, 14.9%). Participants changed their way of interacting very fast: 58 participants (32.0%) changed it after one mistake, and 61 (33.7%) participants after two to three mistakes.

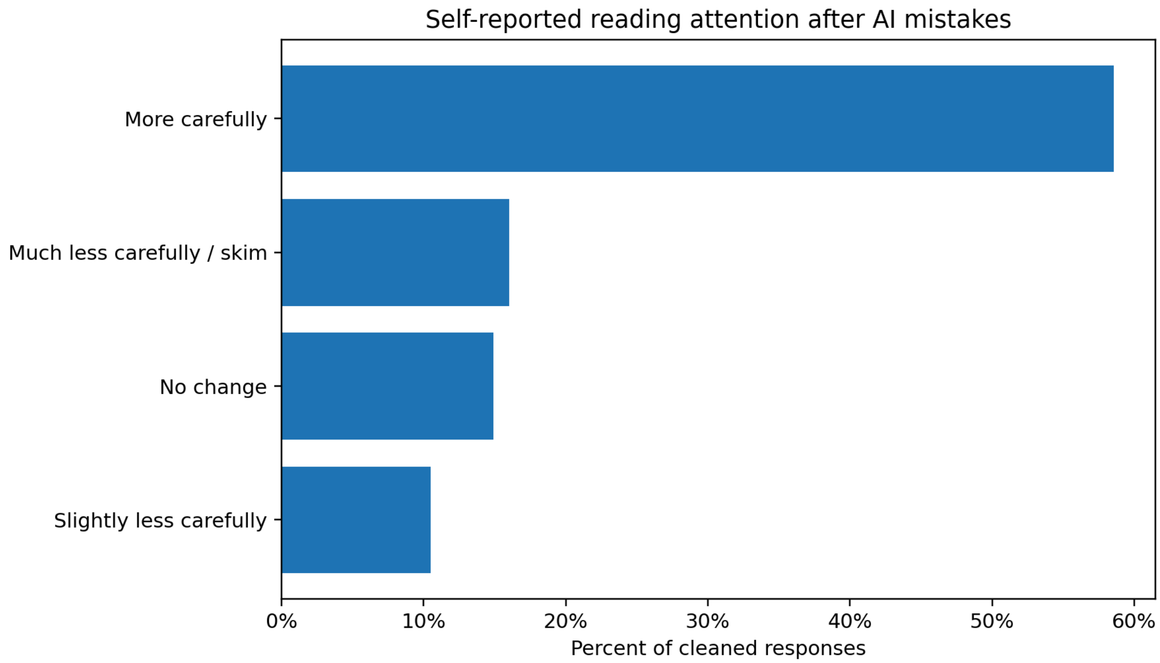


*Figure 4. Self-reported reading-attention changes after AI mistakes.*

## First Action After Repeated Wrong Answers

The most popular reaction to getting repeated wrong answers is trying Google or any other search engine (71, 39.2%), closely followed by rephrasing or simplifying the question (69, 38.1%). Fewer participants tried switching to a different AI (21, 11.6%), giving up (16, 8.8%), or asking the same question again (4, 2.2%). Grouping switch away behaviors, such as switching to Google, switching to a different AI, and giving up, there are 108 respondents (59.7%) who demonstrated it.

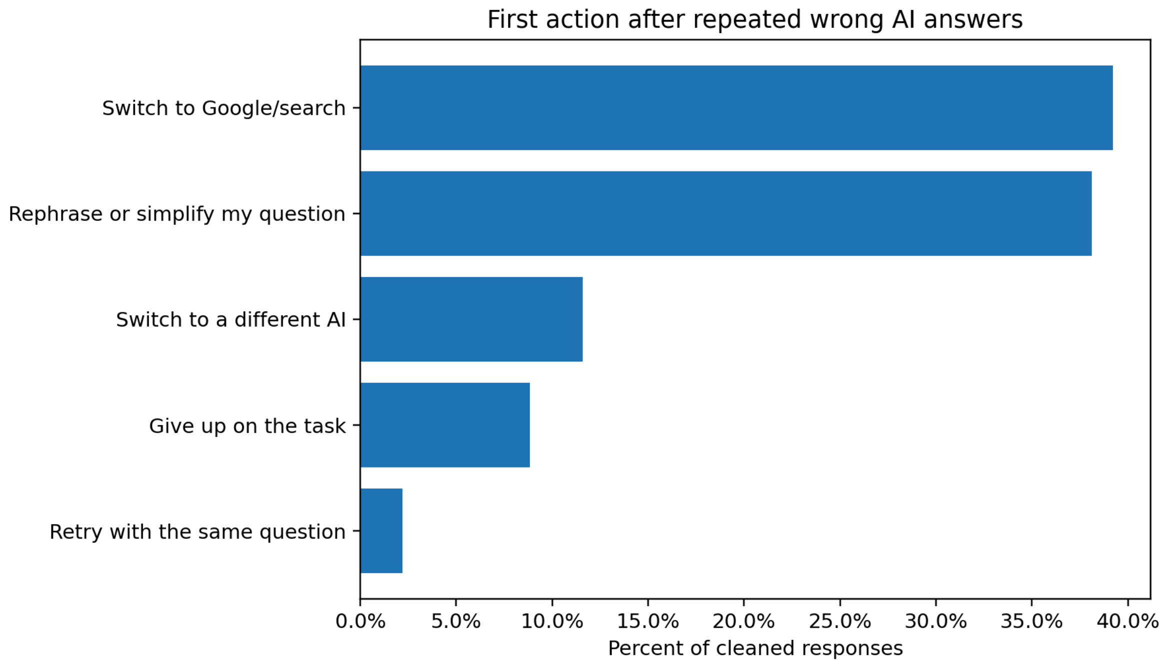


*Figure 5. First action respondents reported taking after repeated wrong AI answers.*

## Frustration Composite

Frustration composite consisting of five items after reverse-coding positively oriented items on the use of time effectively and confidence in the AI has a mean of 4.65 (SD = 1.19). Cronbach's alpha for this five-item scale is adequate for exploratory student surveys ($\alpha = 0.69$).

## Qualitative Themes From Written Responses

The optional written-response question was "What typically causes the participant to become frustrated with mistakes made by the AI chatbot?" In the Google Form summary, there were 37 optional written responses. The written responses were analyzed based on recurring themes instead of being considered as individual statistics. The purpose of the qualitative analysis is to provide additional insight into the patterns seen in the survey.

**Table 4**
**Qualitative themes and illustrative participant quotes**

| Theme | Illustrative quote | Connection to findings |
|---|---|---|
| Verification and citation | *"If I can easily fact check. A good ai puts proper citations to where I can read more. And I often do read more."* | This finding / response from the user connects to the verification shift argument. Some users will check their sources before stopping due to errors. |
| Low expectations (low trust) | *"I generally don't get frustrated as I go in assuming that the answer will be wrong or incomplete..."* | Showcases the role that expectations play in frustration. Users who have a predisposition against artificial intelligence will experience less frustration due to errors. |
| Cost in time and effort | *"Oftentimes, the time saved on what it does right is not made up for by the time lost to fixing its mistakes."* | Links AI failures to the feeling of having one's efforts wasted, especially when users have to become human validators of AI work. |
| Repetitive failures and being unable to correct itself | *"When it doesn't acknowledge its mistakes or change its approach after being called out."* | Explains why repeated failures might lead to greater frustration: users expect that the system will learn from its mistakes and improve. |
| Knowing reason for problem reduces frustration | *"if it's a mistake due to a known and communicated limitation it frustrates me less..."* | Suggests that transparency regarding limitations can help in reducing frustration because failure becomes more understandable. |
| Imposed use and restrictions on access | *"It frustrates me the most when I am forced into it for customer service."* | Demonstrates that frustration is not only about inaccuracy but is enhanced when users feel that there is no other way for them. |

To summarize this, the written responses provided additional insight into the patterns that were observed in the statistics. It seemed like frustration was often caused by wasting time, repeated mistakes, unreliable information sources, and the need for the user to verify information.

# Discussion

## Main Finding

Through these findings, a pattern was observed, in which there was an increase in user frustration due to repeated mistakes by the AI chatbots, but also the more important reason is that the tendency of users changed from trusting the bot to verifying the bot. This change was reflected by three main observations: many participants mentioned that they were paying attention to the AI response, many rephrased their request, and the most popular switch away activity involved changing to Google or another search engine. Such a verification shift is explained by the theory on trust-calibration as users were moving away from automatic reactions to checking activities (Lee & See, 2004).

### Interpretation of H1: Repeated Errors Increased Frustration

H1 was supported. According to the mean frustration levels, it can be noted that it increased as mistakes increased in both scenarios. Scenario A demonstrated a stable upward trend in terms of frustration from the first incorrect response up to the eighth, while Scenario B showed the increase of this variable from early to near-end stages. This finding confirms the findings from HCI research about repeated errors being frustrating since they interfere with goals and require extra recovery actions (Ceaparu et al., 2004; Lazar et al., 2004). Still, according to the results of Scenario A, it should be noted that frustration did not grow at an equal rate for all checkpoints. The biggest increase was noted between the first and third incorrect responses, while the next increase became lower between the fifth and eight errors. In such a way, the frustration starts growing rather fast after the first mistakes and then stops growing and even starts leveling off after users start expecting the errors from the bot. This could be explained by the loss of trust in the bot as more errors occur.

### Interpretation of H2: Inconsistent Errors Were Often as Frustrating as Always-Wrong Errors

H2 was somewhat supported. More participants picked Scenario B over Scenario A as the more frustrating type of failure scenario on its own, but more participants picked the same number for both. This implies that one must not exaggerate these findings in terms of proving that unpredictability was always more frustrating. It is safe to say that inconsistencies were at least as frustrating as always being wrong for a lot of people. Such a result is noteworthy since fewer incorrect responses appeared in Scenario B than in Scenario A, yet more people found Scenario B frustrating. It is linked to the literature on automation since inconsistencies can impede users' ability to create an adequate cognitive representation of how much the automated system can be trusted (Lee & See, 2004; Parasuraman & Riley, 1997).

### Interpretation of H3: Attention Shifted, But Not Always Downward

H3 was supported, but in an unusual way. While the initial hypothesis anticipated that frustration would cause lack of attention and lead the users to skim the texts more carefully, the results revealed that people tended to read AI answers more thoroughly after the mistakes were made. This means that mistakes might make people think skeptically about AI. Instead of becoming less engaged, the respondents became more cautious and started verifying the information provided by the chatbot. In terms of cognitive load theory, careful reading of information can be associated with additional verification activities, not with simple engagement (Mayer, 2005; Sweller, 1988).

### Interpretation of H4: Users Often Repair or Switch Before Giving Up

Similar to H3, the pattern for H4 was more complicated than simple, it was supported although. Few participants stated that they would stop using the service immediately after seeing an error. Instead, respondents tried to fix their question or to use another search tool such as Google. It looks like there is a ladder of behavior in case when AI fails – respondents try to fix the problem or to verify the answer externally first, and only when nothing helps, they switch tools or drop out. The results are consistent with the findings in algorithm aversion literature, where the visible mistakes decrease willingness to use automated advice (Dietvorst et al., 2015).

### The Verification-Shift Argument

This paper is represented in the verification-shift argument. Mistakes made by the chatbot not only are annoying to users, but they additionally change users' approach with AI chatbots. If the chatbot makes a mistake, the user will no longer think about it as the source of correct answers; instead, he or she will begin treating it as a tool that requires verification. In the case of verification, users will perform actions like careful analysis, rephrasing, citation search, Google search, and changing the AI. LLM hallucinations are explained by the research mentioned above (Huang et al., 2023; Ji et al., 2023).

### Role of Expectations

According to survey data, pulling from the written responses, users' expectations may influence the level of frustration experienced towards the chatbot. Some participants reported that they feel less frustrated with AI due to the expectation of the system's inaccuracy or incompleteness and its necessity to verify facts. Other participants reported feeling more frustrated with the system due to the confidence of AI, neglecting the instructions, repetition of the mistake, and reluctance to adjust the strategy despite corrections. This finding means that frustration is caused not only by the inaccuracy of AI, but also by the breach of users' expectations about usefulness, transparency, and control of AI.

### Implications

These findings mean that uncertainty should be communicated, sourcing should be offered, and overconfident answers should be avoided where possible. Implication is consistent with automation-trust literature according to which useful systems should help users form appropriate trust towards it: neither blind dependence nor total mistrust (Parasuraman & Riley, 1997; Lee &

See, 2004). For the perspective of users and students, these results imply that verification skills are as important as prompt-writing skills. For the perspective of researchers, these results imply that frustration should be researched not only emotionally, but also behaviorally.

## Limitations

- Since the experiment relied on scenarios instead of an actual chatbot experience, the data represents imaginary behaviors rather than actual behaviors.
- Attention was self-reported instead of measured using reading time or eye tracking.
- Dropout was assessed using intended action or probable first action and not actual task abandonment.
- The sample size used was a convenient sample and therefore does not represent all AI users.
- The use of the chatbot question allowed for more than one option when it should have been only one since it asked about most common uses.
- There was an inconsistency in the error rates between Scenario A and Scenario B; scenario A presented that all answers were errors whereas Scenario B presents both errors and correct answers.
- Duplicate content responses were also found hence the need to indicate if the final claim is based on the raw data or cleaned data.

## Future Research

The next step researchers should take, branching off from this current research, is to conduct a similar experiment in a simulated environment of live chatbot, where various aspects can be measured such as the time of reading, the length of prompting, the number of retries, fact-checking activities, and actual dropout. Future research may also include comparing various AI platforms, studying the effect of advanced AI users on tolerance of errors and whether citing or placing uncertainty labels may decrease frustration caused by mistakes.

## Conclusion

This study aimed to investigate the relationship between the accumulation of errors committed by AI chatbot and frustration, trust, attention and user behavior. The results obtained indicate the increase of frustration in response to repetition of errors, while the most interesting result is not the drop-out rate. Most of the respondents mentioned that they started being more careful about reading, trying to formulate their questions again, and changing from chatbot to Google or other search engines. Thus, the repetition of errors made the respondents change from trusting interaction with the AI chatbot to verifying interaction with it. The study contributes to the research in the human-AI interaction field, as it demonstrates that user frustration is not only an emotional aspect, but rather related to adaptation and decision making processes in regard to AI usage.

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## Appendix A: Survey Instrument Summary

This survey covered users usages of chatbot platforms, frequency of uses, AI experiences, perceived accuracy of information provided, overall trustworthiness, reduced trustworthiness from answering incorrectly, level of frustration in Scenario A, level of frustration in Scenario B, failure-type comparison, shift in attention to reading, attention shift threshold, first response to repeatedly incorrect answers, a five-item frustration scale, overall reliability of chatbot, overall predictability of chatbot, optional explanations and demographics.

## Appendix B: AI Use Transparency Statement

AI programs have been used for creating outlines, organizing thoughts, drafting, preparing charts and explaining statistics. The student researchers will be held accountable for examining all material, validating the data, rewriting the paper in their own language and ensuring accuracy of statements based on survey results.